Task A: Machine Learning Irrigation Classification Report

AI-Driven Smart Irrigation Recommendation System | Precision Agriculture

1. Research Methodology & Scientific Integrity

This report details the machine learning classification benchmark predicting irrigation requirement (Irrigation_Required). The target is deterministically derived from the FAO-56 hydrological water balance (max(0, ETc - Rain)). Two paradigms are evaluated: Setting A (Upper bound formula reproduction with rainfall) and Setting B (Independent remote sensing using Sentinel-2 MSI reflectance, vegetation indices, ET0, and crop phenology).

2. Model Benchmark Summary Table

Setting	Model	Accuracy	Balanced_Accuracy	Macro_F1	Weighted_F1	NO_Precision	NO_Recall
Baseline	Majority_Class_YES	0.8561	0.5	0.4612	0.7897	0.0	0.0
B_Independent	RandomForest	1.0	1.0	1.0	1.0	1.0	1.0
B_Independent	XGBoost	0.8561	0.5	0.4612	0.7897	0.0	0.0
B_Independent	ExtraTrees	1.0	1.0	1.0	1.0	1.0	1.0
B_Independent	GradientBoosting	0.8561	0.5	0.4612	0.7897	0.0	0.0
B_Independent	ExtraTrees_Tuned	1.0	1.0	1.0	1.0	1.0	1.0
A_FormulaRef	RandomForest	1.0	1.0	1.0	1.0	1.0	1.0
A_FormulaRef	XGBoost	0.8561	0.5	0.4612	0.7897	0.0	0.0
A_FormulaRef	ExtraTrees	1.0	1.0	1.0	1.0	1.0	1.0
A_FormulaRef	GradientBoosting	1.0	1.0	1.0	1.0	1.0	1.0

3. Test Set Confusion Matrix

